

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Kanmani L(192511414) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

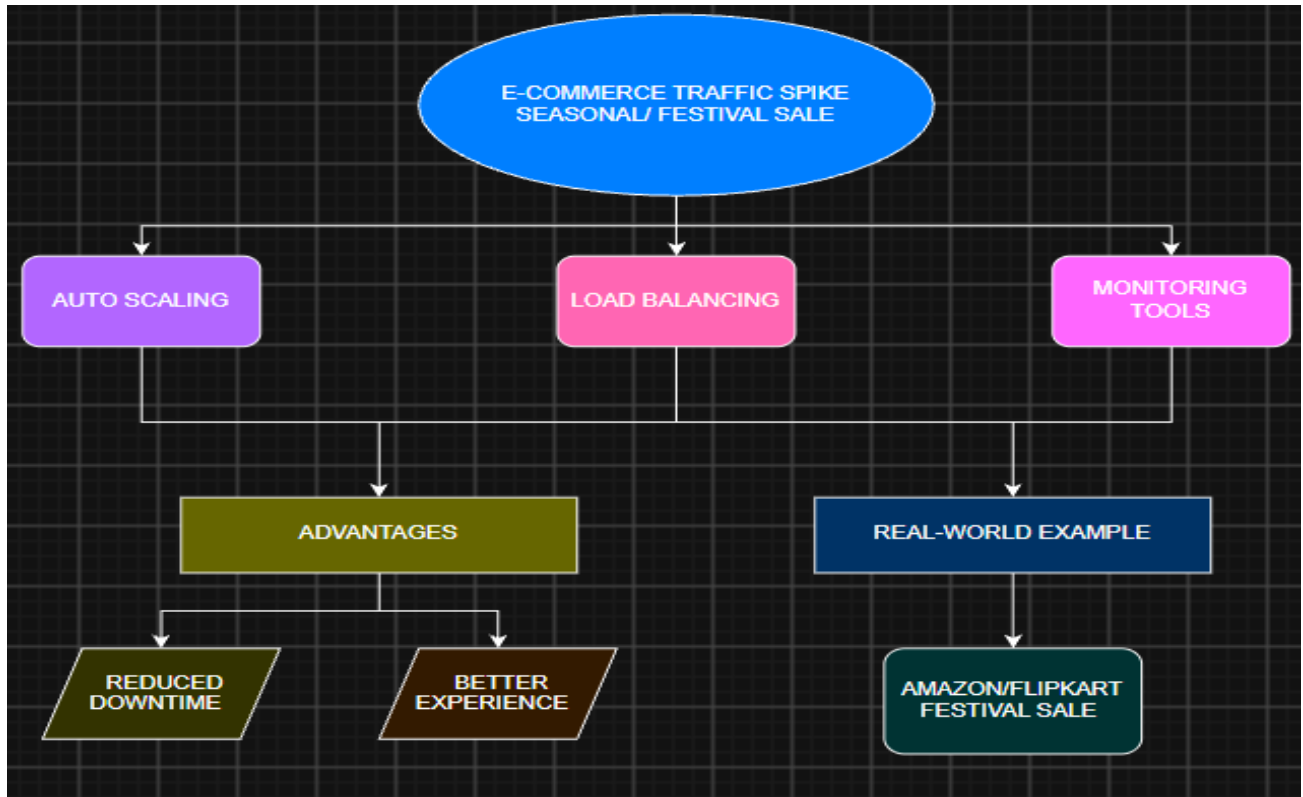
1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer: In an e-commerce company, traffic can increase suddenly during seasonal sales, festival offers, or special events. Cloud auto-scaling and load balancing help the company handle these unpredictable traffic spikes efficiently.

- Auto-scaling: Auto scaling automatically adjusts the number of cloud servers according to the workload. When traffic increases, new virtual machines or instances are added to handle more users. When the traffic decreases, unnecessary instances are removed. This ensures efficient resource utilization and helps reduce cloud costs.
- Load balancing: Load Balancing distributes incoming user requests across multiple available servers.
It prevents a single server from becoming overloaded and ensures that requests are processed efficiently.
If one server fails, the load balancer can redirect users to healthy servers, improving system availability.
- This approach reduces downtime by preventing server overload and providing backup servers during failures. It also improves user experience by providing faster response times, smooth page loading, and uninterrupted services, even when millions of users access the website simultaneously.
- Monitoring tools: Monitoring tools continuously track important metrics such as CPU utilization, memory usage, network traffic, response time, and server health. When resource usage crosses a predefined threshold, monitoring systems can trigger auto-scaling actions. Alerts also help administrators detect failures and performance issues quickly.

Real-world example:

During a major sale such as Amazon's Great Indian Festival or Flipkart's Big Billion Days, the number of users may increase significantly. Auto-scaling adds more servers to handle the extra traffic, while the load balancer distributes customers among these servers. Monitoring tools continuously observe the system and help trigger scaling when required. Thus, the e-commerce platform remains available, reliable, and responsive, resulting in a better customer experience and reduced downtime.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

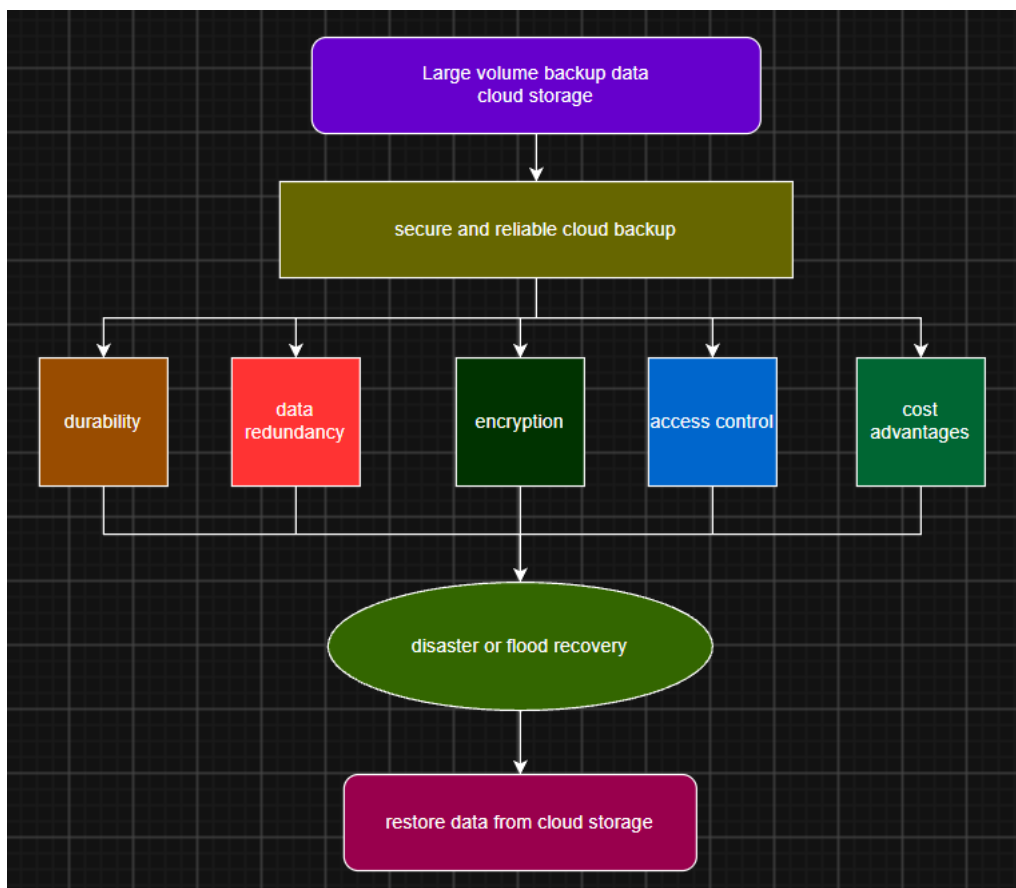
Answer:

Cloud storage provides a secure, scalable, and cost-effective solution for organizations that need to store large volumes of backup data. It ensures that backup data remains available even when hardware failures or disasters occur.

- **Durability and Data Redundancy:** Cloud providers store multiple copies of data across different servers, storage devices, and sometimes geographically separate data centers. This redundancy protects data from hardware failures, accidental deletion, and natural disasters. High durability ensures that the probability of permanent data loss is extremely low.
- **Encryption and Access Control:** Sensitive backup data can be protected using encryption at rest and encryption in transit. Encryption converts data into an unreadable format for unauthorized users.

Access control mechanisms, such as IAM (Identity and Access Management), passwords, multi-factor authentication, and role-based permissions, ensure that only authorized users can access or modify backup data.

- **Cost Advantages:** Cloud storage follows a pay-as-you-go model, so organizations pay mainly for the storage and resources they actually use. There is no need to purchase expensive physical servers, storage devices, or maintain large data centers. Organizations can also use cold or archival storage for infrequently accessed backups at a lower cost.
- **Disaster Recovery Example:** Suppose a company's on-premises data center is damaged by a flood or fire. The company's backup data stored securely in cloud storage can be accessed from another location. The organization can restore its applications and critical data from the cloud backups, reducing downtime and data loss and allowing business operations to continue.



Real-world example: A company can use Amazon S3 for storing backups. S3 provides high durability, encryption, access control, and different storage classes for cost optimization. If the company's local data center is affected by a disaster, the organization can restore its critical data from the cloud backup.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

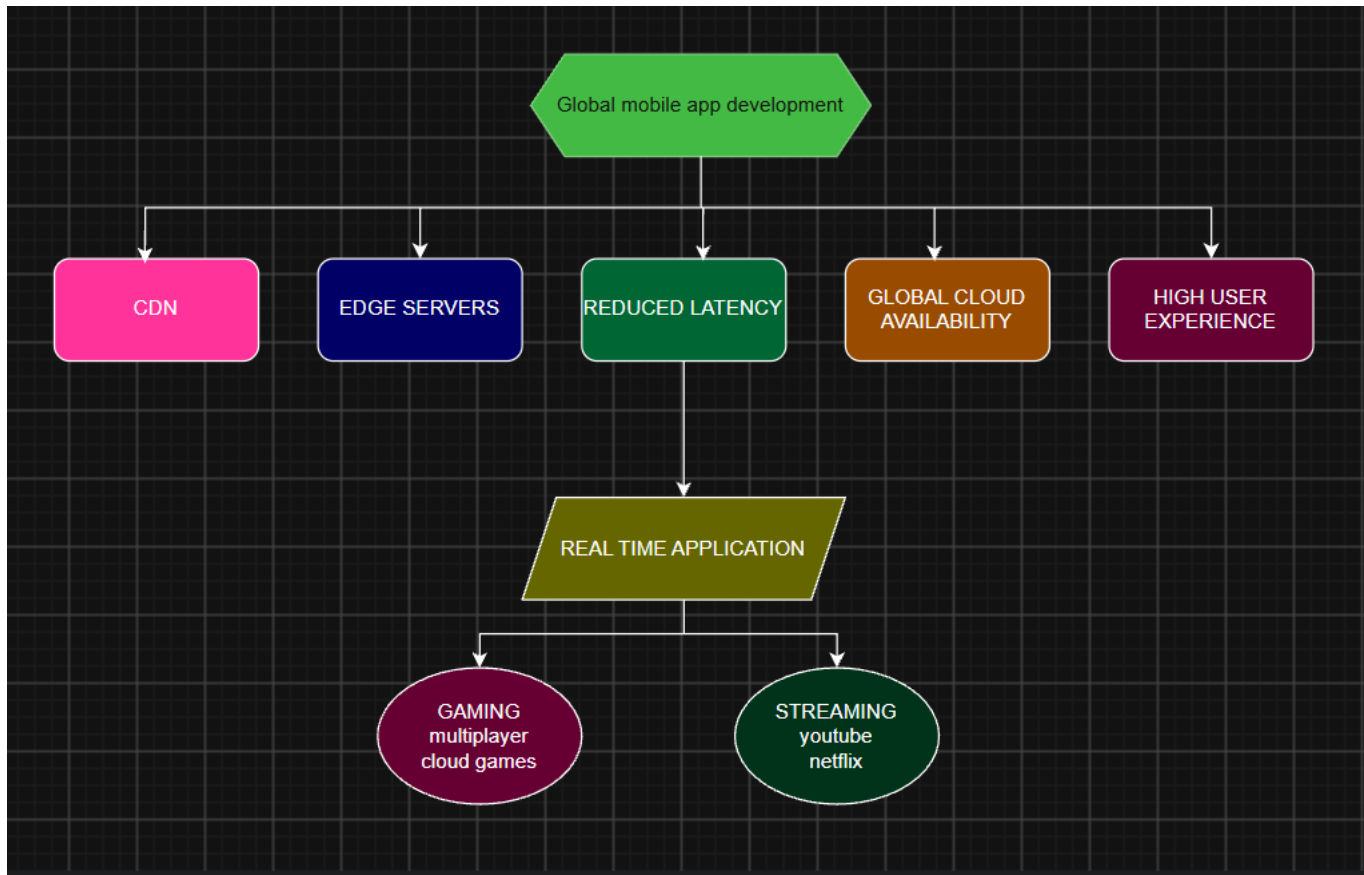
Answer:

When a mobile app is deployed globally, users from different geographical locations may experience delays if all requests are handled by a single central server. Content Delivery Networks (CDNs), edge servers, and global cloud infrastructure help improve application performance and availability.

- **Content Delivery Network (CDN):** A CDN is a distributed network of servers located in different geographical regions. It stores and delivers frequently accessed content such as images, videos, static files, and application resources from a server closer to the user. This reduces the distance data has to travel and therefore reduces latency and improves loading speed.
- **Edge Servers and User Experience:** Edge servers are located close to end users and process or deliver data at the network edge. By serving content from the nearest edge location, they reduce network delay, buffering, and response time. This provides a smoother experience for users across different countries and regions.
- **Global Cloud Availability:** Cloud providers such as AWS, Microsoft Azure, and Google Cloud offer data centers and availability zones across multiple geographic regions. Applications can be deployed across multiple regions, with replication, load balancing, and automatic failover. If one region experiences a failure, traffic can be redirected to another healthy region, ensuring high availability.

Real-time Streaming Example: Consider a global streaming application such as Netflix or YouTube. When a user watches a video, the CDN delivers the content from a nearby edge server instead of a distant central server. This reduces buffering and improves streaming quality. During high demand, cloud infrastructure can automatically scale resources and distribute traffic across regions.

Thus, CDNs reduce latency, edge servers improve responsiveness, and global cloud deployment provides high availability, resulting in a fast and reliable experience for users worldwide.



4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer:

Financial institutions handle highly sensitive information such as customer account details, payment information, and financial transactions. Therefore, they require strong security controls and must comply with regulations and legal requirements while using cloud services.

- **Regulatory Compliance and Auditing:** Cloud providers such as AWS, Microsoft Azure, and Google Cloud provide compliance certifications and security frameworks that help organizations meet regulatory requirements. They offer compliance reports, audit support, and tools to verify whether security policies are being followed. The financial institution is still responsible for configuring its applications and data securely.

- **Logging and Monitoring:** Logging records important activities such as user logins, data access, configuration changes, and financial transactions. Monitoring continuously observes system performance and security events. These tools help detect suspicious activities, unauthorized access, and potential security threats. Logs also provide an audit trail for investigations and regulatory audits.
- **Encryption:** Encryption protects sensitive data by converting it into an unreadable format. Encryption at rest protects stored data, while encryption in transit protects data while it moves between users, applications, and cloud services. Even if unauthorized users access the data, encryption makes it difficult to understand without the required keys.
- **Data Sovereignty:** Data sovereignty means that data is subject to the laws and regulations of the country where it is stored. The company can meet these requirements by selecting appropriate cloud regions and data centers within the required country or legal jurisdiction. It can also use data residency controls, access restrictions, and encryption to ensure that sensitive data is stored and processed according to regulations.

Example: A bank processes an online fund transfer between two customer accounts. The transaction data is encrypted during transmission and while stored in the cloud. Logs record who initiated and approved the transaction, while monitoring systems detect unusual activities such as repeated failed login attempts or suspicious transfers. The bank can store the financial data in a cloud region located within the required country to comply with data sovereignty and financial regulations.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer: Containerization is a method of packaging an application along with its required libraries and dependencies into a lightweight, isolated unit called a container. Containers are widely used for modern application deployment and microservices architectures.

- **Containers vs Virtual Machines:** Virtual machines (VMs) run a complete guest operating system on top of a hypervisor, so they require more memory, storage, and CPU resources. Containers, on the other hand, share the host operating system's kernel and contain only the application and its dependencies. Therefore, containers are lighter, faster to start, and more resource-efficient than VMs. A single host can run many more containers than VMs.

- **Orchestration with Kubernetes:** When an application has many containers, managing them manually becomes difficult. Kubernetes automates container deployment, scheduling, scaling, networking, and management. It can automatically increase or decrease the number of containers based on demand. It also provides self-healing, restarting failed containers and replacing unhealthy ones, which improves application availability.
- **Portability:** Containers package the application and its dependencies together, allowing the same application to run consistently across different environments. A containerized application can be deployed on AWS, Microsoft Azure, Google Cloud, or on-premises infrastructure with minimal changes.

Example: An online food delivery application divided into separate micro services such as User Service, Restaurant Service, Order Service, Payment Service, and Delivery Tracking Service. Each service can run in its own container and be independently deployed or scaled. Kubernetes can manage these containers and automatically scale the Order or Payment services during periods of high demand.

Thus, containers provide lightweight and portable deployment, while Kubernetes simplifies the management, scaling, and availability of containerized applications.

